Schrödinger Equation

Quantum Physics, 3rd Exercise Sheet – Solutions

Exercises

Exercise 1: Evolution of a spin-1/2 in a static magnetic field

Consider a spin-$\frac{1}{2}$ particle in a magnetic field $\mathbf{B}$. The Hamiltonian is

$$H = -\gamma \mathbf{B}\cdot\boldsymbol{\sigma} = \frac{\hbar\omega_L}{2}\mathbf{n}\cdot\boldsymbol{\sigma}, \quad \mathbf{n} = (\sin\theta_B\cos\phi_B,\, \sin\theta_B\sin\phi_B,\, \cos\theta_B).$$

a) Write down the time-dependent Schrödinger equation for $|\psi(t)\rangle$.

Solution. $i\hbar\partial_t\psi(t) = H\psi(t)$ □

b) In case of z directional magnetic field, the Hamiltonian is simply $H = \frac{\hbar\omega_L}{2}\sigma_z$. Calculate the time-evolution with the $|+\rangle = \frac{1}{\sqrt{2}}(|\downarrow\rangle + |\uparrow\rangle)$ initial state. Show that the polarization vector $\mathbf{p} = (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$ precesses around the z axis. */Hint: Find the eigenvectors and the corresponding eigenvalues of Hamiltonian. Then write up the time evolution of the eigenvectors. Finally, use the linearity of the Schrödinger equation: decompose the $|+\rangle$ state as the superposition of the eigenstates, and apply the time-evolution for the eigenstates./*

Solution. Eigenstates of the Hamiltonian: $\psi_\uparrow(t) = e^{-i\omega_L t/2}|\uparrow\rangle$, $\psi_\downarrow(t) = e^{i\omega_L t/2}|\downarrow\rangle$. Using the linearity of the Schrödinger equation:

$$\boxed{\psi(t) = \frac{1}{\sqrt{2}}(e^{i\omega_L t/2}|\downarrow\rangle + e^{-i\omega_L t/2}|\uparrow\rangle)}.$$

The polarization vectors components:

$$p_x(t) = \langle\sigma_x\rangle_{\psi(t)} = \frac{1}{2}\big(e^{i\omega_L t} + e^{-i\omega_L t}\big) = \cos(\omega_L t),$$

$$p_y(t) = \langle\sigma_y\rangle_{\psi(t)} = \frac{1}{2}\big(-ie^{i\omega_L t} + ie^{-i\omega_L t}\big) = \sin(\omega_L t),$$

$$p_z(t) = \langle\sigma_z\rangle_{\psi(t)} = \frac{1}{2}\big(1-1\big) = 0,$$

$$\boxed{\mathbf{p}(t) = \begin{pmatrix}\cos(\omega_L t)\\ \sin(\omega_L t)\\ 0\end{pmatrix}}$$

This is a rotating vector in the xy plane. □

c) Find the eigenstates and eigenenergies of the Hamiltonian with magnetic field in general direction:

$$H = \frac{\hbar\omega_{\mathrm{L}}}{2}\mathbf{n}\cdot\boldsymbol{\sigma}.$$

Use the convention that the first coordinate of the eigenvectors are real. */Hint: For simplicity, solve the eigenvalue equation of $\mathbf{n}\cdot\boldsymbol{\sigma}$ and then just multiply the eigenenergies with the prefactor $\hbar\omega_{\mathrm{L}}/2$. Assume that the eigenstate has the $|\downarrow\rangle + \beta|\uparrow\rangle$ form, then normalize it to one. You may need to use the trigonometric identities:* $\cos(x/2) = \sqrt{\frac{1+\cos x}{2}}$, $\sin(x/2) = \sqrt{\frac{1-\cos x}{2}}$*/*

Solution. Let us find the eigenvectors of $\mathbf{n}\cdot\boldsymbol{\sigma}$:

$$\begin{pmatrix}\cos\theta_B & \sin\theta_B e^{-i\phi_B}\\ \sin\theta_B e^{i\phi_B} & -\cos\theta_B\end{pmatrix}\begin{pmatrix}1\\ \beta_+\end{pmatrix} = \begin{pmatrix}1\\ \beta_+\end{pmatrix} \iff \cos\theta_B + \beta_+\sin\theta_B e^{-i\phi_B} = 1 \iff \tilde{v}_+ = \begin{pmatrix}1\\ \frac{e^{i\phi_B}(1-\cos\theta_B)}{\sin\theta_B}\end{pmatrix}$$

Then normalize it:

$$||\tilde{v}||^2 = 1 + \frac{(1-\cos\theta_B)^2}{\sin^2\theta_B} = 2\frac{1-\cos\theta_B}{\sin^2\theta_B} = \frac{2}{1+\cos\theta_B} \implies |\phi_+\rangle = \begin{pmatrix}\cos(\theta_B/2)\\ e^{i\phi_B}\sin(\theta_B/2)\end{pmatrix}$$

To find the second eigenstate, we search for a state which is orthogonal to $|\phi_+\rangle$ and has a real, negative first component:

$$|\phi_-\rangle = \begin{pmatrix} -\sin(\theta_B/2) \\ e^{i\phi_B}\cos(\theta_B/2) \end{pmatrix}.$$

□

d) Solve the time evolution for an initial state $|\psi(0)\rangle = |\downarrow\rangle = (0,1)^T$ in general magnetic field. */Hint: Use again the decomposition of the initial state to the eigenvectors of the Hamiltonian./*

Solution. We know the time evolution of the eigenstates:

$$|\phi_+(t)\rangle = e^{-i\omega t/2}|\phi_+\rangle, \quad |\phi_-(t)\rangle = e^{i\omega t/2}|\phi_-\rangle$$

The initial state can expanded in the eigenbasis of the Hamiltonian:

$$|\downarrow\rangle = e^{-i\phi_B}\sin(\theta_B/2)|\phi_+\rangle + e^{-i\phi_B}\cos(\theta_B/2)|\phi_-\rangle$$

This implies

$$|\psi(t)\rangle = e^{-i(\omega t/2+\phi_B)}\sin(\theta_B/2)|\phi_+\rangle + e^{i(\omega t/2-\phi_B)}\cos(\theta_B/2)|\phi_-\rangle = \begin{pmatrix} ie^{-i\phi_B}\sin\theta_B\sin(\omega t/2) \\ \cos(\omega t/2) + i\cos\theta_B\sin(\omega t/2) \end{pmatrix}$$

□

e) *Optional:* Show that the evolution corresponds to a rotation of the expectation value of the magnetic moment vector around the direction of $\mathbf{B}$.

Solution. Calculate the time-evolution of the polarization vector.

$$-e^{i\phi_B}\sin\theta_B\sin(\omega t/2)(\cos(\omega t/2) + i\cos\theta_B\sin(\omega t/2)) - e^{-i\phi_B}\sin\theta_B\sin(\omega t/2)(\cos(\omega t/2) - i\cos\theta_B\sin(\omega t/2))$$

$$\begin{aligned} a_x(t) &= \sin\phi_B\sin\theta_B\sin(\omega t) + \cos\phi_B\cos(2\theta_B)\sin^2(\omega t/2), \\ a_y(t) &= -\cos\phi_B\sin\theta_B\sin(\omega t) + \sin\phi_B\sin(2\theta_B)\sin^2(\omega t/2), \\ a_z(t) &= -\cos^2(\omega t/2) - \cos(2\theta_B)\sin^2(\omega t/2) \end{aligned}$$

□

Exercise 2: Electron spin resonance

Consider an electron with spin-$\frac{1}{2}$ in a magnetic field parallel to the z axis. Assume, that the system is in its ground state $|\downarrow\rangle$, and switch on a rotating magnetic field. The system Hamiltonian reads as

$$H(t) = -\gamma\mathbf{B}(t)\cdot\boldsymbol{\sigma} = \frac{\hbar\omega_L}{2}\sigma_z + \frac{\hbar\Omega}{2}(\sigma_x\cos\omega_d t + \sigma_y\sin\omega_d t) \tag{1}$$

a) Consider a time-dependent unitary $U(t)$ and a general Hamiltonian H. Prove the form of the Hamiltonian in the frame defined by U reads as

$$H_U = UHU^\dagger + i\hbar\dot{U}U^\dagger. \tag{2}$$

/Hint: Compute the expression $i\hbar\partial_t|\psi_U(t)\rangle$, where $|\psi_U(t)\rangle = U(t)|\psi(t)\rangle$. Use the Schrödinger equation for the original state

$$i\hbar\partial_t|\psi(t)\rangle = H|\psi(t)\rangle. \tag{3}$$

Finally, find the form of the transformed Hamiltonian from the equation

$$i\hbar\partial_t|\psi_U(t)\rangle = H_U|\psi_U(t)\rangle. \tag{4}$$

/

Solution.

$$i\hbar\partial_t|\psi_U(t)\rangle = i\hbar\partial_t(U(t)|\psi(t)\rangle) = i\hbar\dot{U}(t)|\psi(t)\rangle + i\hbar U(t)\partial_t|\psi(t)\rangle = i\hbar\dot{U}(t)|\psi(t)\rangle + U(t)H|\psi(t)\rangle \tag{5}$$

$$= i\hbar\dot{U}U^\dagger U|\psi(t)\rangle + UHU^\dagger U|\psi(t)\rangle = (i\hbar\dot{U}U^\dagger + UHU^\dagger)|\psi_U(t)\rangle \tag{6}$$

□

b) Transform the system to the rotating frame with the time-dependent unitary $U = e^{i\frac{\omega_d t}{2}\sigma_z}$ and calculate the effective Hamiltonian in the rotating frame: $H_U = UHU^\dagger + i\hbar\dot{U}U^\dagger$.

Solution. Using $e^{i\frac{\omega_d t}{2}} = \cos(\omega_d t/2)I + i\sin(\omega_d t/2)\sigma_z$, and $[\sigma_z, \sigma_x] = 2i\sigma_y$, $[\sigma_z, \sigma_y] = -2i\sigma_x$:

$$e^{i\frac{\omega_d t}{2}\sigma_z}\sigma_x e^{-i\frac{\omega_d t}{2}\sigma_z} = \cos(\omega_d t)\sigma_x - \sin(\omega_d t)\sigma_y, \quad e^{i\frac{\omega_d t}{2}\sigma_z}\sigma_y e^{-i\frac{\omega_d t}{2}\sigma_z} = \cos(\omega_d t)\sigma_y + \sin(\omega_d t)\sigma_x$$

$$UHU^\dagger = \frac{\hbar\omega_L}{2}\sigma_z + \frac{\hbar\Omega}{2}\sigma_x, \quad i\hbar\dot{U}U^\dagger = -\frac{\hbar\omega_d}{2} \implies H_U = \frac{\hbar(\omega_L - \omega_d)}{2}\sigma_z + \frac{\hbar\Omega}{2}\sigma_x$$

□

c) Calculate the time evolution of the system, when the initial state is $|\downarrow\rangle$ in the rotating frame. Calculate the probability of finding the system in state $|\uparrow\rangle$ as a function of time in case of resonance: $\omega_L = \omega_d$.

Solution. In this resonance condition the Hamiltonian is simply $H_U = \frac{\hbar\Omega}{2}\sigma_x$. Then the time-evolution operator is simply

$$U_U(t) = e^{-i\frac{\Omega}{2}\sigma_x} = \cos(\Omega t/2)I - i\sin(\Omega t/2)\sigma_x.$$

Then the time-evolution (in the rotating frame):

$$|\psi_U(t)\rangle = U_U(t)\,|\downarrow\rangle = \cos(\Omega t/2)\,|\downarrow\rangle - i\sin(\Omega t/2)\,|\uparrow\rangle$$

Then, the probability of finding the system in the excited state is

$$p_e(t) = |\,\langle\uparrow|\psi_U(t)\rangle\,|^2 = \sin^2(\Omega t/2) = \frac{1-\cos(\Omega t)}{2}$$

□

d) Transform the solution back to the lab frame. *Homework:* plot the time-evolution of the polarization vector in the lab frame.

Solution.

$$|\psi(t)\rangle_{\text{lab}} = U^\dagger\,|\psi_U(t)\rangle = e^{-i\frac{\omega_d t}{2}\sigma_z}\,|\psi_U(t)\rangle = e^{-i\frac{\omega_d t}{2}}\cos(\Omega t/2)\,|\downarrow\rangle - ie^{i\frac{\omega_d t}{2}}\sin(\Omega t/2)\,|\uparrow\rangle$$

□

e) Solve the time evolution in the rotating frame with nonzero detuning: $\delta = \omega_L - \omega_d$. What is the probability of measuring the system in the excited state as a function of time?

Solution. In finite detuning the rotating frame Hamiltonian is

$$H_U = \frac{\hbar\delta}{2}\sigma_z + \frac{\hbar\Omega}{2}\sigma_x = \frac{\hbar\Omega_{\text{R}}}{2}\left(\frac{\delta}{\Omega_R}\sigma_z + \frac{\Omega}{\Omega_R}\sigma_x\right), \quad \Omega_R = \sqrt{\delta^2 + \Omega^2}.$$

Then the time-evolution operator is

$$U(t) = \cos(\Omega_R t/2) - i\sin(\Omega_R t/2)\left(\frac{\delta}{\Omega_R}\sigma_z + \frac{\Omega}{\Omega_R}\sigma_x\right)$$

Then the time-evolution of the excited state is

$$p_e(t) = \left|\,\langle\uparrow|\cos(\Omega_R t/2) - i\sin(\Omega_R t/2)\left(\frac{\delta}{2\Omega_R}\sigma_z + \frac{\Omega}{\Omega_R}\sigma_x\right)|\downarrow\rangle\,\right|^2 = \frac{\Omega^2}{\Omega_R^2}\sin^2(\Omega_R t/2).$$

□

Exercise 3: Drive of a nitrogen-vacancy center at zero-magnetic field. (homework)

The negatively charged nitrogen-vacancy (NV^-) center NV^- center in diamond consists of a substitutional nitrogen atom adjacent to a vacant lattice site. The static Hamiltonian of the NV^- center ground state in zero magnetic field, neglecting any interactions with nuclear spins or spin-strain interactions, and assuming that the NV^--axis is oriented along the z-direction, is given by

$$H = hDS_z^2, \tag{7}$$

where h is the Planck constant, $D = 2.87$ GHz is the zero-field splitting, and

$$S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{8}$$

is the z component of the dimensionless spin-1 operator. The resulting Hamiltonian has two energy levels: one at zero energy and a doubly degenerate energy level with energy $D = 2.87$ GHz. The system couples to the magnetic field as

$$h\gamma_e \mathbf{S} \cdot \mathbf{B}, \tag{9}$$

where $\gamma_e = 28$ GHZ/T is the electron gyromagnetic ratio, and $\mathbf{S}$ is the vector of the dimensionless spin-1 operators. The x and y components read as

$$S_x = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad S_y = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}. \tag{10}$$

a) Show that the spin-1 operators follow the same commutator relations as the Pauli matrices:

$$[S_x, S_y] = 2iS_z, \quad [S_y, S_z] = 2iS_x, \quad [S_z, S_x] = 2iS_y. \tag{11}$$

(These commutation relations make the Pauli matrices the generators of a representation of the Lie algebra. All spin and angular momentum operators follow the same commutation relations.)

b) Let us apply a resonant magnetic drive of the zero-field system:

$$\mathbf{B} = B \begin{pmatrix} \cos\omega t \\ \sin\omega t \\ 0 \end{pmatrix}, \tag{12}$$

where $\omega/2\pi = D$. To solve the time-dependent Schrödinger equation, apply the time-dependent unitary transformation: $U(t) = e^{i\omega t S_z}$, and obtain the time-independent Hamiltonian of the system in the rotating frame.

$$H_U = UHU^\dagger + i\hbar \dot{U} U^\dagger \tag{13}$$

c) To solve the time-independent Schrödinger equation, let us assume that the magnetic field is weak compared to the zero-field splitting: $\gamma_e B \ll D$. If we completely neglect the effect of the magnetic field, then in the rotating frame there is a doubly degenerate zero energy level and another energy level with $2D$ energy. The weak magnetic field couples the two degenerate levels to each other, and also couples one of the zero energy levels to the high-energy level. Heuristically, neglect the high-energy level and focus on the dynamics inside the degenerate subspace driven by the magnetic field. Solve the dynamics using the truncated, 2×2 effective Hamiltonian, with the initial condition

$$\psi(t = 0) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \tag{14}$$

d) Transform the solution back to the lab frame.

e) Solve the same system with rotating magnetic field in the opposite direction.

f) *Numerical homework exercise.* Check numerically the validity of the heuristic approximation that we applied. Solve the dynamics of the three level system exactly in the rotating frame with ratios: $D/\gamma_e B = 100, 10, 5, 2$. Calculate the overlap between the exact and the approximate solutions.